Automobile Repair & Service History Database

Fixing Service History for Better Business Decision Making

Final Project Report

BUS 385 - Business Data Management

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1. Executive Summary

This project addresses a common challenge in the automobile repair industry: the absence of a centralized system for managing vehicle service histories, customer records, and repair data. Without a centralized system, businesses rely on weak records, manual lookups which take too much time, and disconnected spreadsheets- leading to data inefficiencies and incorrect business decisions. Our solution involved the use of Microsoft Access to create a structured database for a large-scale dataset, and developing SQL queries to support operational and business decisions.

We used a Kaggle dataset containing motor vehicle incidents from 2020 to 2024. The team imported the data into a Microsoft Access table. Using the queries, we performed data retrieval, filtering, aggregation, subqueries, data manipulation, and quality validation. Through this project we turned raw data into valuable business intelligence by using structured data management. Some of our findings were: identifying the most frequent repair types, seasonal service volume trends, customer patterns, and data integrity. The result is a database system which can be used in a real-world auto repair environment.

2. <u>Introduction</u>

2.1 Purpose of the Project

The purpose of this project is to create a fully functioning centralized automobile repair and service history database, addressing the needs of an auto repair business. Project Aims:

- Merge vehicle service records, customer information, and repair details into a single centralized database system.
- Develop SQL queries that make it possible to do fast data retrieval, customer lookups, and inventory awareness.
- Generate business intelligence reports that show repair trends, service frequency, and customer behavior.
- Demonstrate how a structured database decreases data redundancy and increases operational accuracy.

2.2 Relevance to Business Data Management

This project applies concepts we have learned throughout the course. Implementing foundational concepts like field typing, primary key assignment, and constraints in data modeling. In our structured SQL querying, we applied basic SELECT statements to developing sub-queries. Our business scenario presents real-world database challenges that are faced by numerus industries across the world. The use of Microsoft Access also reflects the course's emphasis on using relevant industry-standard tools for business database management.

3. Problem Statement

3.1 Business Problem

Auto repair businesses generate large amounts of service data daily. Each visit produces data from the customer, the vehicle, the type of work done, the parts used, and the overall cost for each action. Without a centralized database, all this data is unorganized and difficult to track.

This fragmentation creates five operational problems:

- Fragmented Service History: Staff members cannot retrieve a vehicle's entire history, leading to wasted time for laborers and customers.
- Disconnected Service Records: Customer contact information is not linked to vehicle ownership records, making it difficult to follow up with customers on service reminders.
- No Parts Traceability: Parts usage is not tracked for specific service sessions, leading to inventory planning being reactive.
- Billing Inefficiencies: Without a central record, generating correct bills and finding unpaid accounts can only be done manually.
- Absence of Business Analytics: Without proper business insights, managers do not have a way to track which services are used the most, which type of vehicles are serviced the most, and how demand shifts according to seasonal changes.

3.2 Data Requirements

To address these challenges, these are the specific data requirements needed:

- Service records show the type of service done, description, and the date of each vehicle visit.
- Customer ID for each service event
- Vehicle type, make, and model
- Towing details like pickup place, drop-off place, and tow date for towing events.
- A structured system that allows quick searches, filtered reports, and overall analysis.

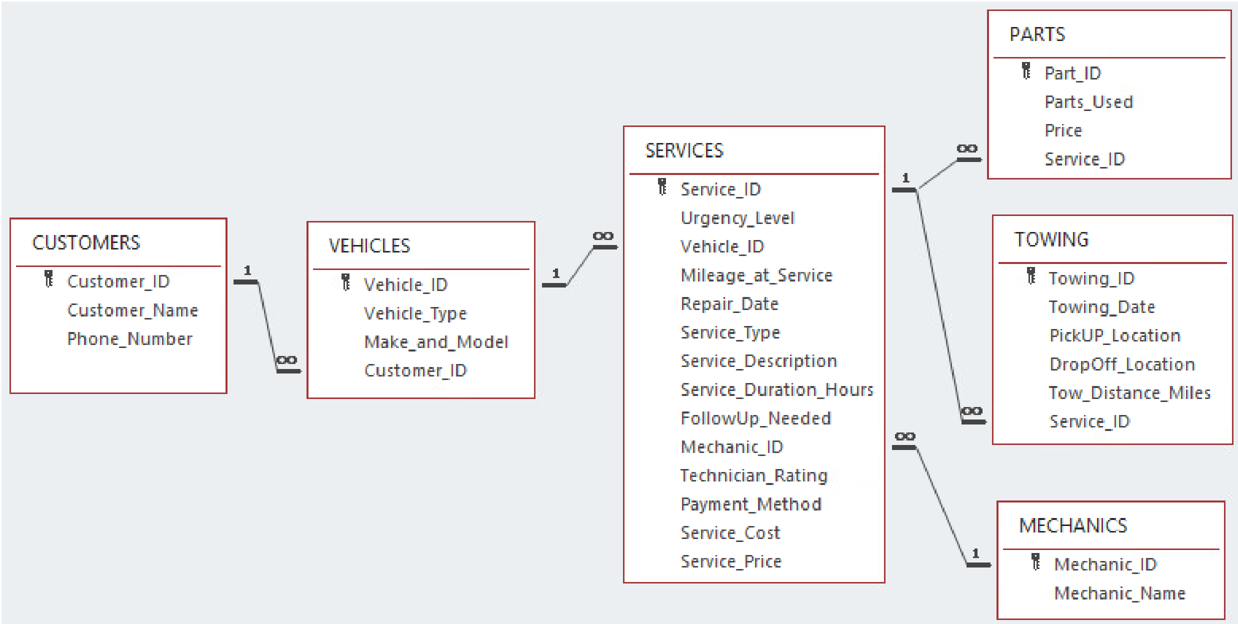
3.3 Business Justification

Addressing this problem is not only technical but also affects the financial and operational aspects of the business. A centralized, structured database for service history helps an auto repair shop to execute lookups quicker, identify important customers for loyalty programs, anticipate parts needed through past repairs, and generate highly accurate financial reports without manual action. These improvements lead to more productivity, better customer retention, and more informed management choices.

4. <u>Database Design</u>

4.1 Entity-Relationship Diagram (ERD)

An Entity-Relationship Diagram summarizes the various relations between different data within a database. This is done in a visual manner, which helps the user see how the various datapoints work with one another. Below is an ERD that summarizes the various relationships between our main data tables.



This ERD showcases the relational connections between our various tables, directly from Microsoft Access. Additionally, there will be one more ERD included as well. Notice the various **one-to-many** connections between our various tables. This is a key part of our database, due to the nature of how our various datapoints interact with one another. This will be discussed further in section 4.3 where we will go over the relation systems for our database.

4.2 Database Schema

CUSTOMERS

Column Name	**Data Type**	**Constraint**	**Description**
Customer_ID	Integer	Primary Key	Unique ID given to each customer
Customer_Name	Short Text		Full name of the customer
Phone_Number	Short Text	NOT NULL	Phone number of the customer

VEHICLES

Column Name	Data Type	Constraint	Description
Vehicle_ID	Long Integer	Primary Key	Unique ID given to each vehicle
Vehicle_Type	Short Text		Category: Car, Truck, Van, SUV, Bus, or Motorcycle
Make_and_Model	Short Text	NOT NULL	Combined make and model
Customer_ID	Integer	Foreign Key	Unique ID given to each customer

SERVICES

Column Name	Data Type	Constraint	Description
Service_ID	Long Integer	Primary Key	Unique ID given to each service event
Urgency_Level	Short Text		Category: Scheduled, Standard, or Emergency
Vehicle_ID	Long Integer	Foreign Key	Unique ID of the vehicle serviced
Mileage_at_Service	Long Integer		Mileage of the vehicle at the time of service
Repair_Date	Date/Time	NOT NULL	Date service was performed
Service_Type	Short Text		One of 7 defined service categories
Service_Description	Long Text	NOT NULL	Description of what was repaired
Service_Duration_Hours	Integer		Amount of hours service took to complete

FollowUp_Needed	Short Text		Yes or No: Does the vehicle need to return
Mechanic_ID	Short Text	Foreign Key	Unique ID of the mechanic who performed the service
Technician_Rating	Integer		Mechanic performance rating out of 5
Payment_Method	Short Text		Type of payment used (Cash, Credit Card, Debit Card, Insurance, or Company Invoice)
Service_Cost	Currency	NOT NULL	The cost of servicing the vehicle
Service_Price	Currency	NOT NULL	Amount charged for completing the service

PARTS

Column Name	**Data Type**	**Constraint**	**Description**
Part_ID	Short Text	Primary Key	Unique ID given to each part
Parts_Used	Short Text	NOT NULL	Type of part
Price	Currency	NOT NULL	Price of the part
Service_ID	Long Integer	Foreign Key	The service ID that the part was used in

TOWING

Column Name	Data Type	Constraint	Description
Towing_ID	Short Text	Primary Key	Unique ID given to each towing event
Towing_Date	Date/Time	NOT NULL	Date of tow
Tow_Location	Short Text		Location the vehicle was picked up from
Drop-off_Location	Short Text		Location the vehicle was dropped off at
Tow_Distance_Miles	Long Integer	NOT NULL	Distance (in miles) between the tow and drop-off location
Service_ID	Long Integer	Foreign Key	ID of the towing event in the SERVICES table

MECHANICS

Column Name	Data Type	Constraint	Description
Mechanic_ID	Short Text	Primary Key	Unique ID given to each mechanic
Mechanic_Name	Short Text	NOT NULL	Full name of the mechanic

4.3 Relationships Between Entities

CUSTOMERS and VEHICLES: This is a one-to-many relationship. One customer can own many vehicles, but each vehicle belongs to only one customer.

VEHICLES and SERVICES: This is a one-to-many relationship. One vehicle can have many service records over time, but each service record is tied to only one vehicle.

SERVICES and PARTS: This is a one-to-many relationship. One service can use many parts, but each parts record belongs to one service.

SERVICES and TOWING: This is a one-to-many relationship. One service can have many towing records linked to it, but each towing record points back to one service.

SERVICES and MECHANICS: This is a many-to-one relationship. Many service records can be assigned to the same mechanic, but each service has only one mechanic listed.

4.4. Design Justification

This database design addresses the auto repair industry's core problem: scattered data across spreadsheets, disconnected records, and a lack of traceability. Consolidating everything into a normalized relational structure ensures data is stored once and referenced everywhere. The SERVICES table acts as the operational hub, linking vehicles, mechanics, parts, and towing into a single quarriable record. This enables business-critical reporting that was previously impossible without manual reconciliation (ex: revenue by service type and technician performance via Technician_Rating). Child tables such as PARTS and TOWING handle one-to-many relationships cleanly (one service, multiple parts; optional towing per job) without bloating the core record. The result is a schema that supports revenue tracking, staff accountability, customer retention, and operational scheduling, all from a single unified system.

5. SQL Queries and Results

All SQL queries were developed and executed in the SQL view of Microsoft Access. This section will present queries from each major category.

5.1 Data Retrieval & Advanced Queries

Query 1 – Service Totals by Brand

```
TRANSFORM COUNT(Service_Type) AS [Count]
SELECT LEFT(Make_and_Model, INSTR(Make_and_Model," ")-1) AS Brand,
       COUNT(VEHICLES.Vehicle_ID) AS [Number of Vehicles]
FROM VEHICLES INNER JOIN SERVICES ON VEHICLES.Vehicle_ID =
SERVICES.Vehicle_ID
GROUP BY LEFT(Make_and_Model, INSTR(Make_and_Model," ")-1)
PIVOT Service_Type;
```

Query Objective: This query analyzes service demand by vehicle brand, breaking down how many times each type of service was performed for different car brands. It helps the business identify the most common types of repairs that each brand requires, supporting targeted maintenance planning, inventory preparation, and service strategy optimization.

Result:

Service Totals By Brand

Brand	Number of Vehicles	Battery Replacement	Brake Service	Engine Repair	Oil Change	Tire Replacement	Towing	Transmission Repair
BMW	1856	272	272	275	263	264	271	239
Chevrolet	1900	265	278	263	296	260	282	256
Dodge	1829	265	276	235	288	243	280	242
Ford	1915	244	268	251	290	293	295	274
Harley-Davidson	1921	285	271	286	268	264	260	287
Honda	1842	227	276	286	281	260	267	245
Toyota	1881	270	269	270	271	261	271	269
Yamaha	1854	248	288	270	244	283	268	253

Query 2 – Average & Total Cost of Parts

```
SELECT Parts_Used AS Parts, COUNT(Parts_Used) AS [Total Used],
        AVG(Price) AS [Average Cost], SUM(Price) AS [Total Cost]
FROM PARTS
GROUP BY Parts_Used
ORDER BY SUM(Price) DESC;
```

Query Objective: This query tracks parts usage and costs by summarizing how frequently each part is used, along with its average and total cost. It helps the business manage inventory, identify high-cost or frequently used parts, and make informed purchasing and cost-control decisions.

Result:

Average & Total Cost of Parts

Parts	Total Used	Average Cost	Total Cost
Clutch Kit	2065	$379.44	$783,542.00
Battery	2076	$208.65	$433,152.00
Oil Pump	1595	$249.48	$397,927.00
Rim	1588	$241.22	$383,051.00
Tire	1561	$179.27	$279,842.00
Calipers	1672	$165.95	$277,468.00
Rotors	1637	$99.51	$162,891.00
Engine Mount	1638	$90.16	$147,689.00
Timing Belt	1574	$89.44	$140,782.00
TPMS Sensor	1643	$64.85	$106,541.00
Transmission Fluid	2065	$49.38	$101,960.00
Brake Pads	1632	$61.90	$101,018.00
Synthetic Oil	2201	$42.46	$93,461.00
Spark Plugs	1601	$55.50	$88,852.00
Battery Holder	2076	$29.86	$61,988.00
Gearbox Seal	2065	$29.81	$61,555.00
Brake Fluid	1653	$20.18	$33,365.00
Oil Filter	2201	$13.86	$30,513.00
Oil Drain Plug	2201	$12.51	$27,538.00
Battery Terminal	2076	$12.49	$25,922.00
Valve Stem	1592	$6.05	$9,638.00

Query 3 – Average & Total Profit by Service

```sql
SELECT Service_Type AS [Service Type], COUNT(*) AS [Total Performed],
        AVG(Service_Price - Service_Cost) AS [Average Profit],
        (SUM(Service_Price) - SUM(Service_Cost)) AS [Total Profit]
FROM SERVICES
GROUP BY Service_Type
ORDER BY AVG(Service_Price - Service_Cost) DESC;
```

Query Objective: This query analyzes the profitability of each service type by calculating how often each service is performed, along with its average and total profit. It helps the business determine which services are most profitable, guiding pricing strategies, resource allocation, and service offerings.

Result:

Average & Total Profit by Service

Service Type	Total Performed	Average Profit	Total Profit
Transmission Repair	2065	$743.48	$1,535,279.00
Towing	2194	$584.46	$1,282,296.00
Engine Repair	2136	$258.40	$551,947.00
Brake Service	2198	$89.02	$195,664.00
Battery Replacement	2076	$53.62	$111,323.00
Tire Replacement	2128	$46.40	$98,738.00
Oil Change	2201	$15.22	$33,502.00

Query 4 – Average Mechanic Ratings

```
SELECT MECHANICS.Mechanic_ID AS ID, Mechanic_Name AS [Mechanic Name],
        ROUND(AVG(Technician_Rating),1) AS [Average Rating / 5]
FROM MECHANICS INNER JOIN SERVICES ON MECHANICS.Mechanic_ID =
SERVICES.Mechanic_ID
GROUP BY MECHANICS.Mechanic_ID, Mechanic_Name
ORDER BY ROUND(AVG(Technician_Rating),1) DESC;
```

Query Objective: This query evaluates mechanic performance by calculating the average customer rating for each mechanic. It helps the business identify top-performing technicians and those who may need improvement, supporting decisions related to training, promotions, and service quality management.

Result:

Average Mechanic Ratings

ID	Mechanic Name	Average Rating / 5
MEC015	Adam Harper	3.1
MEC014	Jerry Allen	3.1
MEC013	Sonny Walker	3
MEC006	Brandon Martinez	3
MEC004	Jamal Ward	3
MEC003	Ryan Weathers	3
MEC001	Miguel Rojas	3
MEC002	Bryan Willis	3
MEC005	Chris Rodriguez	3
MEC012	Hank White	3
MEC011	Willy Adames	3
MEC016	Isaiah Hardy	3
MEC017	Henry Mattews	3
MEC018	Wayne McCormick	3
MEC019	Paul Skens	3
MEC020	Daniel Santos	3
MEC007	Tom Rosseti	2.9
MEC008	Steven Tracey	2.9
MEC010	Bill Walton	2.9
MEC009	Mike Stanton	2.9

Query 5 – Customer Visit & Spending Totals

SELECT Customer_Name **AS** Name, **COUNT**(Service_ID) **AS** [Total Visits],

SUM(Service_Price) **AS** [Total Spent], Phone_Number **AS** [Phone Number]

FROM CUSTOMERS **INNER JOIN** (VEHICLES INNER JOIN SERVICES **ON** VEHICLES.Vehicle_ID = SERVICES.Vehicle_ID) **ON** CUSTOMERS.Customer_ID = VEHICLES.Customer_ID

GROUP BY Customer_Name, Phone_Number

ORDER BY **COUNT**(Service_ID) **DESC**, **SUM**(Service_Price) **DESC**;

Query Objective: This query identifies the most valuable and frequent customers by calculating how many service visits each customer has made and the total amount they have spent. It helps the business recognize top clients, enabling targeted marketing, loyalty programs, and better customer relationship management.

Result:

Customer Visit & Spending Totals

Name	Total Visits	Total Spent	Phone Number
Lincoln Cook	11	$7,108.00	(516)744-6488
Julia Simmons	11	$3,919.00	(516)457-5389
Justin Hayes	10	$7,919.00	(516)807-7876
Luis Simmons	10	$5,121.00	(516)335-2961
Quinn Stevens	10	$5,012.00	(516)989-7967
Gordon Myers	9	$6,530.00	(516)429-5895
Bradley Edwards	9	$5,773.00	(516)360-4951
Chad Morgan	9	$5,069.00	(516)361-7832
Yuri Belvedere	9	$5,047.00	(516)642-2920
Blake Cooper	9	$4,696.00	(516)366-1967
Valeria Lomeli	9	$4,392.00	(516)842-5545
Kevin Adams	9	$4,237.00	(516)707-9144
Stephen Capps	9	$4,154.00	(516)552-3125
Samuel West	9	$4,023.00	(516)595-3930
Jonathan Hayes	9	$4,001.00	(516)620-6428
Fred Baker	9	$3,565.00	(516)485-6270
Tatiana Leyva	9	$2,974.00	(516)744-4133
Noah Bennett	8	$7,687.00	(516)437-5394
Tristan Key	8	$6,144.00	(516)895-3293
Clay Cruz	8	$5,980.00	(516)302-2524
Derek Porter	8	$5,329.00	(516)808-2501
Eva Paiva	8	$5,091.00	(516)817-9252
Reginald Bonilla	8	$5,028.00	(516)701-3938
Roger Hurst	8	$4,946.00	(516)900-4840

Record: 1 of 5011 No Filter Search

Query 6 – Follow-Ups Needed

SELECT *

FROM SERVICES

WHERE FollowUP_Needed = 'Yes';

Query Objective: This query retrieves all service records where a follow-up is required. Its purpose is to help the business identify customers or vehicles that need additional attention, ensuring timely follow-ups, improving customer satisfaction, and preventing unresolved issues from being overlooked.

Result:

Follow-Ups Needed

Service_ID	Urgency_Level	Vehicle_ID	Mileage_at_Service	Repair_Date	Service_Type	Service_Description	Service_Duration_Hours	FollowUp_Needed
111089	Scheduled	211014	99228	12/31/2024	Oil Change	Diagnosed and repaired Synthetic Oil	2	Yes
113643	Scheduled	213540	193056	12/31/2024	Engine Repair	Diagnosed and repaired Spark Plugs	4	Yes
111860	Standard	211779	274609	12/31/2024	Engine Repair	Diagnosed and repaired Oil Pump	5	Yes
103209	Emergency	203200	279466	12/31/2024	Engine Repair	Diagnosed and repaired Timing Belt	4	Yes
106911	Scheduled	206874	83485	12/31/2024	Brake Service	Diagnosed and repaired Calipers	4	Yes
107042	Scheduled	207005	288996	12/31/2024	Transmission Repair	Diagnosed and repaired Gearbox Seal	2	Yes
107559	Scheduled	207516	267315	12/31/2024	Engine Repair	Diagnosed and repaired Oil Pump	4	Yes
105538	Emergency	205512	261329	12/31/2024	Towing	Diagnosed and repaired no parts used	1	Yes
109953	Standard	209889	89410	12/30/2024	Engine Repair	Diagnosed and repaired Spark Plugs	3	Yes
111245	Scheduled	211169	123848	12/30/2024	Transmission Repair	Diagnosed and repaired Clutch Kit	2	Yes
112062	Emergency	211978	250433	12/30/2024	Tire Replacement	Diagnosed and repaired Tire	2	Yes
100109	Scheduled	200109	292748	12/30/2024	Brake Service	Diagnosed and repaired Brake Fluid	1	Yes
111204	Standard	211128	73520	12/29/2024	Engine Repair	Diagnosed and repaired Timing Belt	4	Yes
108011	Emergency	207961	239844	12/29/2024	Engine Repair	Diagnosed and repaired Timing Belt	2	Yes
111670	Standard	211592	278919	12/29/2024	Tire Replacement	Diagnosed and repaired Rim	3	Yes
105911	Standard	205882	187854	12/29/2024	Battery Replacement	Diagnosed and repaired Battery Holder	2	Yes
108378	Standard	208326	84554	12/28/2024	Towing	Diagnosed and repaired no parts used	4	Yes
114250	Scheduled	214137	244424	12/28/2024	Towing	Diagnosed and repaired no parts used	2	Yes
101436	Standard	201436	81320	12/28/2024	Battery Replacement	Diagnosed and repaired Battery Holder	2	Yes
107234	Scheduled	207125	169721	12/28/2024	Engine Repair	Diagnosed and repaired Spark Plugs	2	Yes
106408	Standard	206375	50878	12/28/2024	Engine Repair	Diagnosed and repaired Engine Mount	3	Yes
107155	Standard	207117	265689	12/28/2024	Engine Repair	Diagnosed and repaired Timing Belt	0	Yes
108110	Standard	208060	266840	12/27/2024	Tire Replacement	Diagnosed and repaired TPMS Sensor	4	Yes
110412	Emergency	210343	67537	12/27/2024	Towing	Diagnosed and repaired no parts used	2	Yes

Record: 1 of 7485 No Filter Search

Design View

Query 7 – Insurance Claims

SELECT *

FROM SERVICES

WHERE Payment_Method = 'Insurance';

Query Objective: This query retrieves all service records paid through insurance. Its purpose is to help the business track insurance-based transactions, streamline claims processing, and manage billing or reimbursement more efficiently.

Result:

Insurance Claims

Service_Description	Service_Duration_Hours	FollowUp_Needed	Mechanic_ID	Technician_Rating	Payment_Method	Service_Cost	Service_Price
Diagnosed and repaired Oil Pump	5	Yes	MEC015	1	Insurance	$557.00	$654.00
Diagnosed and repaired Oil Filter	1	No	MEC017	4	Insurance	$78.00	$104.00
Diagnosed and repaired Synthetic Oil	5	No	MEC014	2	Insurance	$49.00	$68.00
Diagnosed and repaired Spark Plugs	3	Yes	MEC015	3	Insurance	$264.00	$782.00
Diagnosed and repaired Clutch Kit	2	Yes	MEC002	4	Insurance	$455.00	$510.00
Diagnosed and repaired Timing Belt	4	Yes	MEC002	3	Insurance	$288.00	$843.00
Diagnosed and repaired no parts used	4	Yes	MEC009	3	Insurance	$150.00	$257.00
Diagnosed and repaired no parts used	2	Yes	MEC010	1	Insurance	$225.00	$1,632.00
Diagnosed and repaired Battery Holder	2	Yes	MEC014	4	Insurance	$251.00	$317.00
Diagnosed and repaired Gearbox Seal	5	No	MEC010	2	Insurance	$334.00	$889.00
Diagnosed and repaired Synthetic Oil	2	Yes	MEC015	3	Insurance	$80.00	$64.00
Diagnosed and repaired Oil Filter	4	Yes	MEC005	1	Insurance	$56.00	$73.00
Diagnosed and repaired Valve Stem	2	Yes	MEC019	4	Insurance	$363.00	$454.00
Diagnosed and repaired Clutch Kit	0	Yes	MEC010	2	Insurance	$622.00	$1,404.00
Diagnosed and repaired Brake Pads	3	No	MEC013	1	Insurance	$245.00	$304.00
Diagnosed and repaired Rotors	4	Yes	MEC006	4	Insurance	$279.00	$334.00
Diagnosed and repaired Tire	6	Yes	MEC016	4	Insurance	$241.00	$328.00
Diagnosed and repaired Oil Drain Plug	5	Yes	MEC017	3	Insurance	$81.00	$67.00
Diagnosed and repaired Oil Pump	1	No	MEC002	5	Insurance	$307.00	$948.00
Diagnosed and repaired Rotors	5	No	MEC020	2	Insurance	$392.00	$398.00
Diagnosed and repaired Battery Terminal	2	No	MEC007	3	Insurance	$233.00	$325.00
Diagnosed and repaired Battery	4	No	MEC009	4	Insurance	$265.00	$363.00
Diagnosed and repaired Battery Holder	2	No	MEC003	3	Insurance	$202.00	$209.00
Diagnosed and repaired Oil Pump	1	Yes	MEC014	2	Insurance	$249.00	$455.00

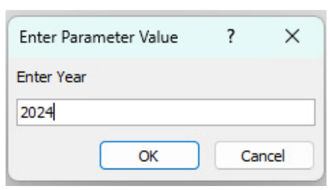
Record: 1 of 2918 No Filter Search

Query 8 – Monthly Profit, Revenue, & Expenses by Year

```sql
SELECT FORMAT(Repair_Date, "mmmm") AS [Month],
        COUNT(Service_ID) AS [Total Services Performed],
        SUM(Service_Cost) AS [Total Expenses], SUM(Service_Price) AS [Total Revenue],
        (SUM(Service_Price) - SUM(Service_Cost)) AS [Total Profit]
FROM SERVICES
WHERE YEAR(Repair_Date) LIKE [Enter Year]
GROUP BY FORMAT(Repair_Date, "mmmm"), MONTH(Repair_Date)
ORDER BY MONTH(Repair_Date);
```

Query Objective: This query summarizes business performance by month for a selected year, showing total services performed, expenses, revenue, and profit. It helps the business analyze seasonal trends, evaluate monthly profitability, and make informed decisions about budgeting, staffing, and operational planning.

Result (2024 is an example, any year that has a service record will work):



Monthly Profit, Revenue, & Expenses by Year

Month	Total Services Performed	Total Expenses	Total Revenue	Total Profit
January	259	$72,234.00	$143,566.00	$71,332.00
February	216	$63,471.00	$121,396.00	$57,925.00
March	250	$71,750.00	$149,026.00	$77,276.00
April	256	$70,724.00	$133,923.00	$63,199.00
May	255	$67,396.00	$127,779.00	$60,383.00
June	275	$73,244.00	$143,374.00	$70,130.00
July	263	$77,534.00	$144,850.00	$67,316.00
August	247	$73,470.00	$137,266.00	$63,796.00
September	252	$69,424.00	$126,312.00	$56,888.00
October	244	$71,220.00	$125,899.00	$54,679.00
November	235	$65,802.00	$124,527.00	$58,725.00
December	265	$74,730.00	$143,497.00	$68,767.00

Query 9 – Towing Mileage Information

```
SELECT ROUND(AVG(Tow_Distance_Miles),1) AS [Average Towing Distance],
        MIN(Tow_Distance_Miles) AS [Lowest Towing Distance],
        MAX(Tow_Distance_Miles) AS [Highest Towing Distance],
        ROUND((SUM(Service_Cost) / SUM(Tow_Distance_Miles)),2) AS [Average Cost/Mile]
FROM TOWING INNER JOIN SERVICES ON TOWING.Service_ID = SERVICES.Service_ID;
```

Query Objective: This query analyzes towing operations by calculating the average, minimum, and maximum towing distances, as well as the average cost per mile. It helps the business evaluate towing efficiency, control costs, and make data-driven decisions on pricing, routing, and resource allocation for towing services.

Result:

Towing Mileage Information

Average Towing Distance	Lowest Towing Distance	Highest Towing Distance	Average Cost/Mile
53.1	5	100	3.67

Query 10 – Vehicles Serviced More Than Once

SELECT Make_and_Model **AS** Vehicle, VEHICLES.Vehicle_ID **AS** [Vehicle ID],

Customer_Name **AS** Owner, Phone_Number **AS** [Phone Number],

COUNT(Service_ID) **AS** [Total Services]

FROM CUSTOMERS **INNER JOIN** (VEHICLES **INNER JOIN** SERVICES **ON** VEHICLES.Vehicle_ID = SERVICES.Vehicle_ID) **ON** CUSTOMERS.Customer_ID = VEHICLES.Customer_ID

GROUP BY Customer_Name, Phone_Number, Make_and_Model, VEHICLES.Vehicle_ID

HAVING COUNT(Service_ID) > 1

ORDER BY COUNT(Service_ID) DESC , Customer_Name;

Query Objective: This query identifies vehicles that have received multiple services, showing the owners information, vehicle details, and the total number of services. It helps the business spot repeat repairs, which may indicate ongoing mechanical issues, allowing for better diagnostics, customer follow-up, and potential warranty or maintenance recommendations.

Result:

Vehicles Serviced More Than Once

Vehicle	Vehicle ID	Owner	Phone Number	Total Services
Toyota CR-V	204138	Anna Freeman	(516)414-9284	3
Harley-Davidson RAV4	202892	Aiden Sullivan	(516)401-3491	2
BMW School Bus	203639	Aimee Corcoran	(516)788-9335	2
Ford Sportster	203913	Albert Mendez	(516)693-7863	2
Toyota YZF-R6	204477	Alec Donovan	(516)559-7697	2
Honda Coach	206889	Allen Fields	(516)642-2811	2
Toyota F-150	208367	Allen Ward	(516)323-1070	2
Chevrolet Odyssey	201060	Amanda Carter	(516)646-4837	2
Ford Sienna	203312	Andres Molina	(516)812-4011	2
BMW Coach	201644	Angel Torres	(516)521-1226	2
Harley-Davidson Transit	207848	Angelique Rivas	(516)355-9377	2
Honda Odyssey	209309	Anika Schaeffer	(516)668-3303	2
Yamaha Transit	204681	Anthony Wells	(516)437-6377	2
Honda Ram 1500	205953	Arianna Long	(516)800-3303	2
Yamaha Coach	200304	Aurora Diaz	(516)298-1741	2
Dodge School Bus	200389	Austin Parker	(516)873-5356	2
Yamaha Coach	201322	Austin Snyder	(516)708-1355	2
Chevrolet School Bus	210254	Avery Lewis	(516)583-6011	2
Toyota Ninja ZX-14R	207095	Barbara Hayes	(516)938-7410	2
Chevrolet Coach	200217	Bobbie Plummer	(516)761-1548	2
Yamaha Mustang	205543	Brandon Carter	(516)231-6596	2
Yamaha YZF-R6	200716	Brent Mills	(516)468-3473	2
Toyota Mustang	201502	Brooks Hughes	(516)494-8139	2
Chevrolet Coach	204712	Cameron Dunn	(516)913-9794	2

Record: 1 of 122 No Filter Search

Query 11 – Customer Records Search

SELECT Customer_Name **AS** Name, V.Vehicle_ID **AS** [Vehicle ID],

Make_and_Model **AS** Vehicle, Service_Type **AS** Service,

Repair_Date **AS** [Repair Date], Service_Price **AS** Price

FROM CUSTOMERS **AS** C **INNER JOIN** (VEHICLES **AS** V **INNER JOIN** SERVICES **AS** S **ON** V.Vehicle_ID = S.Vehicle_ID) **ON** C.Customer_ID = V.Customer_ID

WHERE Customer_Name **LIKE** "" & [Enter Customer Name] & ""

ORDER BY Repair_Date **DESC**

UNION ALL SELECT "","","","","Total Spent", **SUM**(sub.Price)

FROM (

SELECT Customer_Name, V.Vehicle_ID, Make_and_Model, Service_Type, Repair_Date, Service_Price **AS** Price

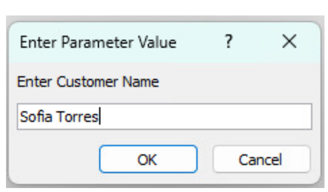
FROM CUSTOMERS **AS** C **INNER JOIN** (VEHICLES **AS** V **INNER JOIN** SERVICES **AS** S **ON** V.Vehicle_ID = S.Vehicle_ID) **ON** C.Customer_ID = V.Customer_ID

WHERE Customer_Name **LIKE** "*" & [Enter Customer Name] & "*"

) **AS** sub;

Query Objective: This query allows users to search for a specific customer's service history by entering their name, returning their vehicles, services performed, repair dates, and service costs. It helps the business quickly access customer records for support, billing inquiries, and personalized service.

Result (Sofia Torres is an example, any name in the database will work):



Customer Records Search

Name	Vehicle ID	Vehicle	Service	Repair Date	Price
Sofia Torres	200002	Harley-Davidson Ram 1500	Tire Replacement	7/8/2020	$636.00
Sofia Torres	204425	Toyota Sienna	Brake Service	2/23/2024	$366.00
Sofia Torres	208794	Yamaha Silverado	Transmission Repair	7/7/2023	$1,071.00
Sofia Torres	210711	Toyota Odyssey	Brake Service	7/9/2020	$446.00
Sofia Torres	213252	Ford Sportster	Engine Repair	5/8/2022	$421.00
				Total Spent	$2,940.00

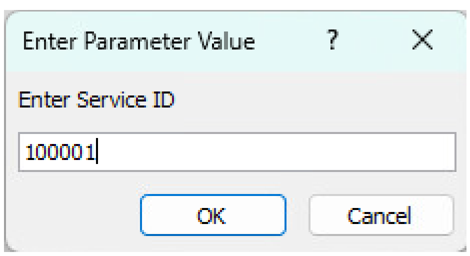
Query 12 – Parts Used For Each Service

```sql
SELECT S.Service_ID AS [Service ID], Service_Type AS Service,
        Repair_Date AS [Repair Date], Parts_Used AS [Parts Used], P.Price
FROM SERVICES AS S INNER JOIN PARTS AS P ON S.Service_ID = P.Service_ID
WHERE S.Service_ID LIKE [Enter Service ID]


UNION ALL SELECT  ' ', ' ', ' ', 'Total Cost', SUM(sub.Price)
FROM (
SELECT S.Service_ID, Service_Type, Repair_Date, Parts_Used, P.Price
FROM SERVICES AS S INNER JOIN PARTS AS P ON S.Service_ID = P.Service_ID
WHERE S.Service_ID LIKE [Enter Service ID]
) AS sub;
```

Query Objective: This query allows users to search for a specific service by entering a Service ID, returning the parts used and the total cost of parts for every repair. It helps the business track repair composition, monitor costs, and ensure accurate billing and transparency in service records.

Result (100001 is an example, any Service ID will work):



Parts Used For Each Service

Service ID	Service	Repair Date	Parts Used	Price
100001	Oil Change	5/14/2022	Oil Filter	$19.00
100001	Oil Change	5/14/2022	Oil Drain Plug	$11.00
100001	Oil Change	5/14/2022	Synthetic Oil	$31.00
			Total Cost	$61.00

5.2 Data Manipulation – INSERT, UPDATE, DELETE

Adding a new customer:

```
INSERT INTO CUSTOMERS
(Customer_ID,Customer_Name,Phone_Number)
VALUES (11498, 'Joshua Hayes', '(516)878-8141');
```

Result: A new customer row has been created.

Adding a new vehicle for an existing customer:

```
INSERT INTO VEHICLES (Vehicle_ID, Vehicle_Type, Make_and_Model,
Customer_ID)
VALUES (210466, 'Van', 'Ford Sienna', 1);
```

Result: A new vehicle record is created with a customer ID of 1.

Updating a customer phone number:

```
UPDATE CUSTOMERS
SET Phone_Number = ' (516)421-2413'
WHERE Customer_ID = 5;
```

Result: The phone number for Customer_ID 5 has been updated.

Deleting a duplicate parts record:

```
DELETE FROM PARTS
WHERE Part_ID = 'B9999';
```

Result: The parts record labeled B9999 is removed.

6. Data Analysis and Interpretation

6.1 Profit Analysis by Service Type

When we examined the data to find the total money earned from each type of service, we found some valuable insights. Transmission Repair has brought in $1,535,279 – which is nearly half of the shop's entire profit. Towing also added $1,282,296 to the overall profit. Both these services make up nearly 75% of all the money the shop makes, even though they make up less than 30% of the total jobs performed.

Furthermore, Oil Changes are the most common job done in the shop, but Oil Changes only bring in $33,502 to the total profit. This analysis shows us that the shop makes the most profit from major and complex jobs, not the routine ones. If the shop looks to grow its profits, they should make sure to hire staff and buy equipment to handle more transmission and towing work, since these jobs generate more money.

6.2 Service Volume and Annual Trends

Looking at the number of services done each year, the numbers are consistent. From 2020 to 2024, the shop performed around 3,000 services each year. But there was a slight drop in 2021 and 2022, which makes sense because there were parts and supply shortages many auto shops faced in that pandemic period. By 2023 and 2024, the numbers went back up to 3,000 services done each year.

This consistency in annual trends is good news for the business. It means that the shop can confidently plan, without worrying about swinging demands.

6.3 Vehicle Type Distribution

The database has six vehicle types: Cars, Trucks, SUVs, Vans, Buses, and Motorcycles. Each type of vehicle makes up roughly 16% to 17% of all vehicles in the system. The most common vehicles are Trucks at 2,517, and the least common are Motorcycles at 2,417, but the difference is small.

This even spread means the business must focus on all types of vehicles. The shop requires valuable tools, equipment, and mechanics for all six vehicle types. The fact that the business repairs Buses, Trucks, and Motorcycles prove the shop serves a mix of commercial customers and personal vehicle owners.

6.4 Follow-Up and Insurance Patterns

Almost half of all service records are marked as needed for a follow-up. That is 7,485 out of 14,998 records, without a database it would be nearly impossible to track all of this. With the centralized database we created, the shop can pull up the complete follow-up list withing seconds, with each customer's phone number listed.

2,918 service records involved an insurance claim. Every one of those insurance claims needs proper documentation and record-keeping to get paid correctly. A centralized database makes that process more reliable and efficient than paper records.

6.5 Towing Patterns

Across the entire dataset, the shop dealt with 2,194 towing events. Pickups were spread evenly across six location types – Residential, Dealer, Garage, Highway, Commercial, and Parking Lot. The average towing distance is 53.1 miles, while the shortest is 5 miles and the longest is 100 miles.

The even spread of pickup locations shows us that the towing service covers a wide area rather than being concentrated in one location. This is very useful information for the business to know how many tow trucks the shop needs and where they should be positioned to respond quickly.

7. Challenges & Solutions

Relations & Connections – Due to the vast size of our database, figuring out how to connect the various tables took quite a bit of effort. One specific issue we had was deciding where to connect to the MECHANICS table. At first, we connected this table to the VEHICLES table, but this was something that didn't make much logical sense. In the end, we had to ensure that the MECHANICS table was connected through the SERVICES table, to efficiently create a **one-to-many** relationship, where many services can be applied to the same mechanic, which narrows the focus of the MECHANICS table to its primary purpose. This system, in a way, almost breaks our database into distinct groups, such as the first initial link between the CUSTOMERS and VEHICLES tables. These two initial tables focus solely on the customer and their rightful vehicles. The VEHICLES table then flows into the SERVICES table, which focuses on the actual work done on the customer's vehicle, which is the key business offering of this entire system. The

SERVICES table then flows into the PARTS, TOWING, and MECHANICS tables, which all can be summarized as singular events under the broader umbrella of vehicle servicing. The determination of this data flow took a bit of experimentation and logical thinking, but in the end, it provides the most accurate presentation of the data, which is crucial for query development, which we will discuss further.

Queries – Queries are specific commands a user can send to a database to extract valuable information. This is one of the most crucial components of database management. What makes queries tedious is that they are essentially endless in how many things you can do with them, so you must narrow your vision in a way, to bring forth the best information possible when creating queries. For us, the most important queries revolved around revenue, vehicle statistics, customer visits, mechanic ratings, and the usage of parts. These are all key concepts in running a business, because they revolve around revenue, quality of work, and customer retention. These queries almost all revolve around JOINS and aggregate functions. Take for example, our monthly profit queries. These queries require us to take financial information from our SERVICES table, organize them by month, and then aggregate them by Service_Cost, Service_Price, and Profit, to present the data in a clean, income statement-oriented manner. Determining what information is necessary, alongside the syntax to extract the information effectively takes time, but it’s a crucial statement in professional database design.

Data Consistency – Maintaining data consistency across our various related tables was an important part of the creation of this database. All these tables are connected to one another, so keeping the data whole is extremely important. Due to this, it was important to maintain that all foreign keys are connected to their correct primary keys in an accurate manner. This ensures the prevention of potential orphan records, such as a service linked to a vehicle that doesn’t exist. Through this diligence, we can ensure that each service links to a valid vehicle, and each vehicle links to a valid customer, trickling backwards in our data flow, which ensures integrity.

7.1 Microsoft Access Scale Limitations

Our original Kaggle dataset was extremely large, with well over a million different entries. Microsoft Access struggled with the size of the dataset, so to make it more manageable, we cut the dataset down to around 15,000 entries. Regardless, all of this can be transferred from Access to something more robust, such as MySQL Workbench, powered by MySQL. There would be some adjustments necessary for such a transfer, such as data type differences, but the data will still transfer cleanly, at least 90% of the way, with all our work within Microsoft Access.

7.2 Synthetic Data Inconsistencies

One point to note is the use of synthetic data. Synthetic data can, at times, be unrealistic and logically inconsistent. Due to this nature, synthetic data should be approached with a grain of salt, but this does not violate the structure of this database. Even in the use of synthetic data, we made sure all the various tables connected and worked with one another in a logical, acceptable, and useful manner. Even if the data isn't rooted in real world systems, we ensured that our synthetic data was clean and as realistic as possible, which highlights the importance of data validation, even when working with synthetic datasets. This provides the benefit of implementing the structure of the database to real life instances of similar issues, without major changes necessary.

8. Recommendations

8.1 Database management practices

Based on our project findings, there are some recommendations that can be suggested for improving the database managements practice in the automobile repair industry. One recommendation would be to incorporate more analytics and automation of data. As seen in the database, there was some analytical data such as seasonal demand shifts, customer return rate, and less frequent repair jobs. These types of analytical data can use automated reports which would help with demand forecasting. By using automated data, businesses will have a better predictive analysis of what parts/inventory to store, and this would help with inventory problems such as inventory shortages or surpluses. This can also help with labor forecasting as the business can rely on the automated data to anticipate when they should increase or decrease their workforce to meet customer demands.

Another recommendation that can be suggested would be to integrate the data with other systems such as accounting software. Linking these systems would streamline a lot of jobs such as invoice and payment history by automatically transferring the data without the need to input manual entries. This would lead to a decrease in data duplication/data inaccuracy. Less billing errors would also be a beneficial result as well. Integrating the system to things such as online appointment scheduling can also streamline a lot of processes for businesses as well. Automated messaging systems confirming appointment dates along with follow up messages can help increase customer satisfaction and also help with retaining their customer base. Connecting systems such as these would reduce the amount of administrative work needed, improve data accuracy, while improving customer satisfaction as well.

8.2 Future Iterations

In future iterations of this project, the use of data warehousing, big data concepts, and advanced SQL features can enhance business intelligence capabilities. A data warehouse can be created to store tremendous amounts of data such as customer history, repair history, and inventory. By creating a data warehouse, business owners can track long-term analytics such as revenue growth and returning customer history. Big data tools can also be utilized if the business expands to multiple locations. By having a larger dataset, it can provide predictive insights such as future repair likeliness and regional demand patterns which would help the business by being better prepared for what's to be expected. To add on, in future iterations of the project, more advanced SQL features can be used to improve the performance of the database. Features such as triggers, indexing, and stored procedures can help increase the efficiency of the database.

9. Conclusion

9.1 Key Takeaways

The completion of the project came with a few key takeaways for our team. One of the key takeaways that we realized through the project is the value of how beneficial a centralized database can be for a business. By combining multiple aspects such as vehicle information, vehicle history, and customer history, the business can operate in a much more efficient manner. By combining information, it reduces the likeliness of data duplication and data inaccuracy. This also makes it faster to search for customer information which streamlines day-to-day operations. With Microsoft Access, we were able to design tables, create relationships, and manage data while SQL aided with updating and retrieving data through queries.

9.2 Reflection

This project was an insightful practice of all the materials we learned in class. Through this project we were able to incorporate all the different lessons and apply it into a real-world business example. To complete the project, we had to incorporate core database designs principles such as entity relationships, primary keys, normalization, foreign keys, etc. Designing the relationships between customers, vehicles parts, etc. also gave us experience in creating a database structure that would be used in a business setting. This project also allowed us to practice our technical skills in SQL. We were able to retrieve and manipulate data by using SELECT, JOIN, INSERT, UPDATE, and DELETE statements that we had learned in class. In addition, working with Microsoft Access gave us insight on potential challenges that can occur, such as data limitations. In conclusion, this project was an invaluable experience for our team as we were able to incorporate things that we learned from class and we were able to practice firsthand in a real-world business type of setting.

10. Appendices

Raw Data: RAW vehicle_repair_data.xlsx

Clean Data: CLEAN & MANIPULATED vehicle_repair_data

References

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